Raydium Protocol Architecture

On-chain programs, shared primitives and the off-chain surface that fronts them

LEGEND

- Call / CPI
- Legacy / inert path
- SDK — Builds instructions (client-side)
- TX API — Builds instructions (server-side)
- AGG — Direct CPI by aggregators
- IDX — Indexed by Raydium indexer

1 / USER ENTRY POINTS — who calls Raydium

Out of scope: Perps
Orderly Network integration, not a Raydium program

Raydium web UI
raydium.io

Custom apps
SDK or direct CPI

External aggregators
Jupiter, 1inch, etc.

2 / OFF-CHAIN SURFACE — composes the programs; none of it is on-chain

REST API
api-v3.raydium.io
Read-mostly cache + quote engine
Does not build transactions

quotes

TypeScript SDK
@raydium-io/raydium-sdk-v2
Client-side transaction builder
Reads fresh state from RPC; knows every account layout

quotes

Transaction API
transaction-v1.raydium.io
Server-side builder: returns
ready-to-sign transactions

reads

Indexers & price feeds
Parse program logs into reserves,
24h volume and APR

schema

IDL registry
raydium-io/raydium-idl
Anchor IDLs: schema source of truth

Instructions are composed into ONE Solana transaction (N instructions)
No orchestrator program owns the flow. Badges on each program show who touches it.

3 / ON-CHAIN PROGRAMS — Solana mainnet-beta; independent programs sharing conventions

Peers, not a stack
AMM programs never call each other. Only AMM Routing composes several pools, via CPI, in a single transaction.

AMM Routing
multi-pool router; CPIs into all 4 AMMs
TX API · AGG

LP-Lock is a wrapper
Holds LP under a PDA so fees stay claimable but liquidity cannot be withdrawn. It is not a fifth AMM.

CPI per hop

AMM v4
constant-product; the original AMM
SDK · IDX

CPMM — RECOMMENDED
constant-product, Token-2022 native
SDK · TX API · IDX

CLMM
concentrated liquidity (Uniswap v3 style)
SDK · TX API · IDX

Stable AMM
lookup-table curve; router-only pairs
IDX

AMM PROGRAMS: four independent pool formats

MigrateToAmm (legacy state)

MigrateToCpswap

LaunchLab
bonding-curve launches; graduate to CPMM
SDK · IDX

LP-Lock / Burn & Earn
wraps CPMM and CLMM LP positions
SDK

Farm v6
v3 / v5 are wind-down only
SDK

inert

Every program CPIs into these

4 / SOLANA PLATFORM PRIMITIVES — shared by every program

OpenBook v2
CLOB; AMM v4 wiring is inert

SPL Token + Token-2022
token transfers via CPI

System / Rent / Associated Token
account creation, rent, user ATAs

Source: docs.raydium.io/protocol-overview/architecture | Unofficial redesign for readability; program IDs live in reference/program-addresses.